Suggested Answers for Seminar Items

Assessment objective 3

Item one:

To identify which copper compound is responsible for leaching into natural water bodies and causing systemic toxicity, we evaluate the enthalpy of solution for both compounds using Born-Haber cycles and Hess's Law

A. Anhydrous Copper(II) Chloride ($CuCl_2$)

1 Lattice Formation Enthalpy ($\Delta H^{\ominus}_{lat}$):

$$\Delta H^{\ominus}_f(CuCl_2) = \Delta H^{\ominus}_{at}(Cu) + IE_1(Cu) + IE_2(Cu) + 2\Delta H^{\ominus}_{at}(Cl$$

$$-220 = +338 + 745 + 1958 + 2(121) + 2(-349) + \Delta H^{\ominus}_{lat}$$

$$-220 = 2585 + \Delta H^{\ominus}_{lat} \implies \Delta H^{\ominus}_{lat} = -2805 \text{ kJ mol}^{-1}$$

Enthalpy of Solution ($\Delta H^{\ominus}_{sol}$):

$$\Delta H^{\ominus}_{sol} = -\Delta H^{\ominus}_{lat} + \Delta H^{\ominus}_{hyd}(Cu^{2+}) + 2\Delta H^{\ominus}_{hyd}(Cl^-)$$

$$\Delta H^{\ominus}_{sol} = +2805 + (-2100) + 2(-360) = -15 \text{ kJ mol}^{-1}$$

Conclusion for $CuCl_2$: The solution process is exothermic ($\Delta H^{\ominus}_{sol} < 0$), making $CuCl_2$ highly soluble in water.

B. Copper(II) Sulfide (CuS)

1 Enthalpy of Solution ($\Delta H^{\ominus}_{sol}$):

$$\Delta H^{\ominus}_{sol} = \text{Lattice Dissociation Energy} + \Delta H^{\ominus}_{hyd}(Cu^{2+}) + \Delta H^{\ominus}_{hyd}$$

$$\Delta H^{\ominus}_{sol} = +3506 + (-2100) + (-1050) = +356 \text{ kJ mol}^{-1}$$

- Conclusion for CuS: The solution process is endothermic ($\Delta H^{\ominus}_{sol} \gg 0$) due to its very high lattice energy ($+3506 \text{ kJ mol}^{-1}$). This makes CuS insoluble in water.

- Source of Poisoning: Anhydrous Copper(II) Chloride ($CuCl_2$) dissolves significantly in runoff water, releasing free toxic copper ions ($Cu^{2+}_{(aq)}$) into the marshland.

- Insoluble Salt Safety: Copper(II) sulfide (CuS) remains insoluble in natural water systems and poses minimal immediate systemic danger.

Suggested Answers for Seminar Items

Assessment objective 3

Item one:

To identify which copper compound is responsible for leaching into natural water bodies and causing systemic toxicity, we evaluate the enthalpy of solution for both compounds using Born-Haber cycles and Hess's Law

A. Anhydrous Copper(II) Chloride ($CuCl_2$)

1 Lattice Formation Enthalpy ($\Delta H^{\ominus}_{lat}$):

$$\Delta H^{\ominus}_{f}(CuCl_2) = \Delta H^{\ominus}_{at}(Cu) + IE_1(Cu) + IE_2(Cu) + 2\Delta H^{\ominus}_{at}(Cl$$

$$-220 = +338 + 745 + 1958 + 2(121) + 2(-349) + \Delta H^{\ominus}_{lat}$$

$$-220 = 2585 + \Delta H^{\ominus}_{lat} \implies \Delta H^{\ominus}_{lat} = -2805 \text{ kJ mol}^{-1}$$

Enthalpy of Solution ($\Delta H^{\ominus}_{sol}$):

$$\Delta H^{\ominus}_{sol} = -\Delta H^{\ominus}_{lat} + \Delta H^{\ominus}_{hyd}(Cu^{2+}) + 2\Delta H^{\ominus}_{hyd}(Cl^{-})$$

$$\Delta H^{\ominus}_{sol} = +2805 + (-2100) + 2(-360) = \mathbf{-15 \text{ kJ mol}^{-1}}$$

Conclusion for $CuCl_2$: The solution process is exothermic ($\Delta H^{\ominus}_{sol} < 0$), making $CuCl_2$ highly soluble in water.

B. Copper(II) Sulfide (CuS)

1 Enthalpy of Solution ($\Delta H^{\ominus}_{sol}$):

$$\Delta H^{\ominus}_{sol} = \text{Lattice Dissociation Energy} + \Delta H^{\ominus}_{hyd}(Cu^{2+}) + \Delta H^{\ominus}_{hyc}$$

$$\Delta H^{\ominus}_{sol} = +3506 + (-2100) + (-1050) = \mathbf{+356 \text{ kJ mol}^{-1}}$$

- Conclusion for CuS: The solution process is endothermic ($\Delta H^{\ominus}_{sol} \gg 0$) due to its very high lattice energy ($+3506 \text{ kJ mol}^{-1}$). This makes CuS insoluble in water.

- Source of Poisoning: Anhydrous Copper(II) Chloride ($CuCl_2$) dissolves significantly in runoff water, releasing free toxic copper ions ($Cu^{2+}_{(aq)}$) into the marshland.

- Insoluble Salt Safety: Copper(II) sulfide (CuS) remains insoluble in natural water systems and poses minimal immediate systemic danger.

To establish effective administration for EDTA therapy, we analyze the elimination kinetics in human plasma using Data Set II

Time t (hours)	[EDTA] (mol dm ^{3})	ln[EDTA]
1.0	1.48	0.3920
2.0	1.45	0.3716
3.0	1.43	0.3577
4.0	1.41	0.3436
5.0	1.38	0.3221
6.0	1.36	0.3075

$$\ln[\text{EDTA}]_t = \ln[\text{EDTA}]_0 - k \cdot t$$

Plotting ln[EDTA] against time yields a straight line with a constant negative slope confirming **First-Order Elimination Kinetics**

For first-order elimination kinetics:

$$t_{1/2} = \frac{\ln(2)}{k} = \frac{0.69315}{0.01672\ \text{hr}^{-1}} \approx \mathbf{41.5}\ \text{hours}$$

Rate Constant (k):

$$k = -\frac{\ln(1.36) - \ln(1.48)}{6.0 - 1.0} = -\frac{0.3075 - 0.3920}{5.0} \approx \mathbf{0.0167}\ \text{hr}^{-1}$$

The balance between lattice energy and hydration enthalpy determines whether an ionic salt dissolves in water. Highly exothermic solution enthalpies allow salts to dissolve which contaminate rivers, wetlands, and drinking water. Therefore, Treat the industrial runoff with salts that will lead to the precipitation of the toxic ions

EDTA follow first-order elimination kinetics, where the rate of drug clearance is directly proportional to its plasma concentration. Establishing the rate constant and half-life allows medical personnel to maintain effective drug concentration levels in the body without exceeding toxic limits. Therefore if a drug has a longer half-life, the dosage should be spaced such that it does not accumulate in the body to toxic levels.

Item two

1. Enthalpy of Formation ($\Delta H_f^\ominus$) of ZnO

Born–Haber cycle components:

- $\Delta H_{at}^\ominus(\mathrm{Zn}) = +126\ \mathrm{kJ\ mol^{-1}}$
- $\mathrm{IE}_1(\mathrm{Zn}) = +906\ \mathrm{kJ\ mol^{-1}}$
- $\mathrm{IE}_2(\mathrm{Zn}) = +1733\ \mathrm{kJ\ mol^{-1}}$
- $\frac{1}{2}E(\mathrm{O=O}) = +249\ \mathrm{kJ\ mol^{-1}}$
- $\mathrm{EA}_1(\mathrm{O}) = -142\ \mathrm{kJ\ mol^{-1}}$
- $\mathrm{EA}_2(\mathrm{O}) = +731\ \mathrm{kJ\ mol^{-1}}$
- $U_L(\mathrm{ZnO_{formation}}) = -4010\ \mathrm{kJ\ mol^{-1}}$

$$\Delta H_f^\ominus(\mathrm{ZnO}) = 126 + 906 + 1733 + 249 - 142 + 731 - 4010$$

$$\mathbf{\Delta H_f^\ominus(ZnO) = -407\ kJ\ mol^{-1}}$$

2. Displacement Enthalpy ($\Delta H_{rxn}^\ominus$)

$$\mathrm{ZnO_{(s)} + M_{(s)} \rightarrow MO_{(s)} + Zn_{(s)}}$$

$$\Delta H_{rxn}^\ominus = \Delta H_f^\ominus(\mathrm{MO}) - \Delta H_f^\ominus(\mathrm{ZnO})$$

$$\Delta H_{rxn}^\ominus = -602 - (-407)$$

$$\mathbf{\Delta H_{rxn}^\ominus = -195\ kJ\ mol^{-1}}$$

Cell Voltage Calculation ($E_{cell}^\ominus$)

$$\Delta H_{rxn}^\ominus = -nFE_{cell}^\ominus \quad (n = 2, F = 96{,}500\ \mathrm{C\ mol^{-1}})$$

$$E_{cell}^\ominus = -\frac{-195{,}000}{2 \times 96{,}500} = \mathbf{+1.01\ V}$$

The calculated cell voltage is **below the target**.
Lattice energy and formation enthalpy help to determine the thermal and electrochemical stability of materials in batteries. To achieve a target voltage greater than 1.20V, pair zinc with a metal system whose oxide formation enthalpy is significantly more negative than -638.6 kJ mol-1, or utilize standard half-cell potential tables to choose couple combinations with a net electromotive force exceeding 1.20V.

Item three

1. Determination of Reaction Orders & Rate Equation

The general rate law for the reaction $2\mathrm{X}_{(aq)} + \mathrm{Y}_{(aq)} \rightarrow \text{Products} + \mathrm{H}_{2(g)}$ is:

$$\text{Rate} = k[\mathrm{X}]^m[\mathrm{Y}]^n$$

Order with respect to X (m):

Comparing **Experiment 1** and **Experiment 2** (where $[\mathrm{Y}]$ is constant at $0.10\ \text{mol dm}^{-3}$):

$$\frac{\text{Rate}_2}{\text{Rate}_1} = \frac{8.0 \times 10^{-3}}{2.0 \times 10^{-3}} = 4 = \left(\frac{0.20}{0.10}\right)^m = 2^m \implies m = 2$$

Order with respect to Y (n):

Comparing **Experiment 1** and **Experiment 4** (where $[\mathrm{X}]$ is constant at $0.10\ \text{mol dm}^{-3}$):

$$\frac{\text{Rate}_4}{\text{Rate}_1} = \frac{4.0 \times 10^{-3}}{2.0 \times 10^{-3}} = 2 = \left(\frac{0.20}{0.10}\right)^n = 2^n \implies n = 1$$

Overall Rate Equation & Order:

$$\text{Rate} = k[\mathrm{X}]^2[\mathrm{Y}] \quad (\text{Overall order} = 2 + 1 = 3)$$

2. Calculation of Rate Constant (k)

Using data from **Experiment 1**:

$$k = \frac{\text{Rate}}{[\mathrm{X}]^2[\mathrm{Y}]} = \frac{2.0 \times 10^{-3}\ \text{mol dm}^{-3}\ \text{s}^{-1}}{(0.10\ \text{mol dm}^{-3})^2 \times (0.10\ \text{mol dm}^{-3})} = \mathbf{2.0}$$

3. Enthalpy Change of Reaction ($\Delta H^{\ominus}_{\text{rxn}}$)

Using standard enthalpies of formation ($\Delta H^{\ominus}_{f}$), noting that $\Delta H^{\ominus}_{f}(\text{H}_{2(g)}) = 0\ \text{kJ mol}^{-1}$:

$$\Delta H^{\ominus}_{\text{rxn}} = \sum \Delta H^{\ominus}_{f}(\text{Products}) - \left[2 \times \Delta H^{\ominus}_{f}(\text{X}) + \Delta H^{\ominus}_{f}(\text{Y})\right]$$

$$\Delta H^{\ominus}_{\text{rxn}} = -670 - [2(-120) + (-250)] = \mathbf{-180\ \text{kJ mol}^{-1}}$$

- **Criterion 1 (Rate Constant):** Calculated $k = 2.0\ \text{mol}^{-2}\ \text{dm}^{6}\ \text{s}^{-1} > 2.0 \times 10^{-2}$ threshold $\rightarrow$ **Satisfied.**
- **Criterion 2 (Thermodynamic Exothermicity):** Calculated $\Delta H^{\ominus}_{\text{rxn}} = -180\ \text{kJ mol}^{-1} < -150\ \text{kJ mol}^{-1}$ threshold $\rightarrow$ **Satisfied.**
- **Verdict:** The reaction **meets all company efficiency criteria** and is **highly suitable** for electricity generation and commercial hydrogen production.

Part (b): Factors Influencing Fuel Production Rate

1. **Reactant Concentration:** Increasing $[\text{X}]$ or $[\text{Y}]$ increases collision frequency per unit time, accelerating hydrogen gas output (with a doubled concentration of X having a 4-fold rate effect due to second-order kinetics).
2. **Temperature:** Raising the temperature increases the kinetic energy of reactant particles, raising the fraction of collisions exceeding activation energy (E_a).
3. **Catalyst Addition:** Introducing a suitable catalyst provides an alternative pathway with a lower activation energy, significantly increasing the rate of hydrogen generation.

Exothermic reactions that generate enable clean off-grid power generation, directly reducing dependence on carbon-emitting fossil fuels.

Determining reaction orders and the rate constant allows chemical engineers to precisely control conditions for optimal hydrogen yield.

Properties of a Good Fuel

1. **High Calorific Value:** Releases a large amount of heat energy per unit mass or volume during combustion.
2. **Moderate / Controlled Ignition Temperature:** Ignores and ignites easily when needed, but not so low that it creates spontaneous combustion hazards during storage.
3. **Moderate Rate of Combustion:** Burns smoothly and steadily at a controlled rate rather than exploding instantaneously or burning sluggishly.
4. **Low Moisture Content:** Contains minimal water content so that heat energy is spent producing power rather than evaporating internal moisture.
5. **Low Non-Combustible / Ash Content:** Leaves minimal solid residue (ash or clinker) after burning, reducing furnace clogging and disposal effort.
6. **Low Toxicity & Minimal Environmental Impact:** Produces non-corrosive, non-toxic flue gases with low harmful emissions (such as minimal SO_x, NO_x, carbon monoxide, or soot).
7. **Safe & Easy Storage and Transport:** Can be stored safely over long periods without significant degradation, leakage, or high risk of accidental explosion.
8. **High Availability & Cost Efficiency:** Readily available in required quantities at a reasonable cost.

Answers to Assessment Objective four

Item one

Reason for Freezing Point Depression:

Adding non-volatile naphthalene disrupts the ability of benzene molecules to form an ordered solid crystal lattice. This lowers the vapor pressure of the solvent, requiring a lower temperature to reach equilibrium between the solid and liquid phases.

Limitations of Cryoscopy for Molecular Mass Determination:

- The solute must be non-volatile, thermally stable, and must not dissociate or associate in the solvent.
- Precise temperature measuring equipment is required, as small measurement errors lead to large discrepancies in calculated molar mass
- Strictly valid only for highly dilute ideal solutions.

Calculation of Relative Molecular Mass (M_r):

$$\Delta T_f = T_f^\circ - T_f = 5.50^\circ\text{C} - 5.216^\circ\text{C} = 0.284^\circ\text{C}$$

Using $\Delta T_f = K_f \cdot m$ where $K_f = 5.12^\circ\text{C kg mol}^{-1}$:

$$m = \frac{0.284}{5.12} \approx 0.05547 \text{ mol kg}^{-1}$$

$$\text{Moles of solute} = 0.05547 \text{ mol kg}^{-1} \times 0.02025 \text{ kg} = 0.001123$$

$$M_r = \frac{0.142 \text{ g}}{0.001123 \text{ mol}} = \mathbf{126.42} \text{ g mol}^{-1}$$

2. Chemical Equilibrium Constant (K_c) & Elimination Conditions

- **Reaction:**

$$\text{C}_6\text{H}_{6(l)} + 3\text{H}_{2(g)} \rightleftharpoons \text{C}_6\text{H}_{12(l)} \quad (\Delta H = -208 \text{ kJ mol}^{-1})$$

- **Equilibrium Quantities:**
  - Initial $\text{C}_6\text{H}_6 = \frac{7.8 \text{ g}}{78 \text{ g mol}^{-1}} = 0.10 \text{ mol}$
  - Initial $\text{H}_2 = \frac{0.6 \text{ g}}{2 \text{ g mol}^{-1}} = 0.30 \text{ mol}$
  - Unreacted $\text{C}_6\text{H}_6 = 0.08 \text{ mol} \implies 0.02 \text{ mol reacted}$
  - H_2 reacted $= 3 \times 0.02 = 0.06 \text{ mol} \implies$ Remaining $\text{H}_2 = 0.24 \text{ mol}$
  - $[\text{H}_2]_{\text{eq}} = \frac{0.24 \text{ mol}}{2.0 \text{ dm}^3} = 0.12 \text{ mol dm}^{-3}$
- **Equilibrium Constant (K_c):**

Since pure liquids have an activity of 1:

$$K_c = \frac{1}{[\text{H}_{2(g)}]^3} = \frac{1}{(0.12)^3} = \mathbf{578.70} \text{ mol}^{-3} \text{ dm}^9$$

Operational Conditions to Maximize Benzene Elimination:

1 **Low Temperature:** Lowering the temperature shifts the equilibrium to the right because the forward reaction is exothermic ($\Delta H = -208\ \text{kJ mol}^{-1}$).

2 **High Pressure:** Increasing pressure shifts equilibrium toward the side with fewer gas moles ($3\ \text{moles}\ \text{H}_2 \rightarrow 0\ \text{gas moles}$).

3 **Excess Hydrogen Gas:** Injecting excess H_2 drives complete conversion of toxic benzene into cyclohexane.

Applications of Freezing Point Depression (Colligative Properties)

- **De-Icing Roads and Aircraft:** Salt is spread on icy roads in winter to lower the freezing point of water, melting ice even at sub-zero temperatures. Similarly, ethylene glycol mixtures are sprayed onto aircraft wings before takeoff to prevent ice formation during flight.
- **Automobile Engine Antifreeze/Coolants:** Radiator fluids use a mixture of water and ethylene glycol. This depresses the freezing point of water (preventing engine block cracking in freezing temperatures) while simultaneously elevating its boiling point (preventing overheating in warm weather).
- **Cryoscopic Determination of Molecular Masses:** In pharmaceutical research and chemical quality control, colligative property measurements are used to verify the molecular weight and purity of synthesized organic compounds.
- **Food Preservation & Frozen Foods:** Controlling the concentration of dissolved solutes (sugars, salts, and alcohols) in frozen desserts (like ice cream) lowers the freezing point, ensuring a smooth, soft texture by preventing large ice crystals from forming.

Applications of Chemical Equilibrium Industrial Synthesis of Chemical Feedstocks:

- **Haber-Process:** Synthesizes ammonia from nitrogen and hydrogen under high pressure and moderate temperature to optimize yield for fertilizer manufacturing.
- **Contact Process:** Converts sulfur dioxide to sulfur trioxide using vanadium (V) oxide catalyst and optimized temperature/pressure conditions to produce sulfuric acid.

Item Two

1. Determination of Chemical Identity of Fumes (Q) and Substance (P)

Empirical & Molecular Formula Calculation:

$$\text{Mass of Sulfur (S)} = 50.05\ \text{g}$$

$$\text{Mass of Oxygen (O)} = 100.00\ \text{g} - 50.05\ \text{g} = 49.95\ \text{g}$$

$$\text{Moles of S} = \frac{50.05}{32.06} = 1.561\ \text{mol}$$

$$\text{Moles of O} = \frac{49.95}{16.00} = 3.122\ \text{mol}$$

$$\text{Mole Ratio (O : S)} = \frac{3.122}{1.561} = 2:1 \implies \text{Empirical Formula} = \text{SO}_2$$

With a molar mass of $64\ \text{g mol}^{-1}$, Gas Q is Sulfur Dioxide (SO_2).

Equilibrium Conversion Reaction:

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$

2. Quantitative Equilibrium Analysis at $400^\circ C$ and $500^\circ C$

At $400^\circ C$:

Initial: 3.0 mol SO_2, 1.5 mol O_2

Equilibrium $O_2 = 0.5$ mol $\implies$ Reacted $O_2 = 1.0$ mol

Equilibrium $SO_2 = 3.0 - 2(1.0) = \mathbf{1.0}$ mol

Equilibrium $SO_3 = 2(1.0) = \mathbf{2.0}$ mol

Percent Conversion: $\frac{2.0}{3.0} \times 100\% = \mathbf{66.7\%}$

At $500^\circ C$:

Equilibrium $SO_2 = 40\%$ of 3.0 mol $= \mathbf{1.2}$ mol

Reacted $SO_2 = 3.0 - 1.2 = 1.8$ mol

Equilibrium $O_2 = 1.5 - \frac{1.8}{2} = \mathbf{0.6}$ mol

Equilibrium $SO_3 = \mathbf{1.8}$ mol

Percent Conversion: $\frac{1.8}{3.0} \times 100\% = \mathbf{60.0\%}$

3. Factors to Facilitate Conversion of Toxic Fumes ($SO_2 \rightarrow SO_3$)

1. Operating Temperature (Exothermic Enthalpy Effect):

Increasing temperature from $400^\circ C$ to $500^\circ C$ decreased SO_3 yield from 66.7% to 60.0%. This proves the forward conversion is **exothermic** ($\Delta H < 0$). Lowering operating temperature favors higher conversion, though a catalyst (V_2O_5) at $400–450^\circ C$ is used to ensure a sufficient reaction rate.

2. Increased Pressure:

The reaction involves a reduction in gas volume (3 moles of gas $\rightarrow$ 2 moles of gas). Applying higher pressure shifts the equilibrium to the right, maximizing SO_3 formation.

3. Excess Oxygen Feed:

Adding an excess stoichiometric ratio of O_2 drives the conversion of residual SO_2 forward (Le Chatelier's principle).

Application explained: This oxidation step forms the core stage of the **Contact Process**, producing sulphuric acid for agricultural phosphate fertilizers, detergents, and lead-acid batteries. This is because dilute sulphuric acid is an electrolyte. It is used as a fertilizer because it contains the sulphate ion that is good for plant growth

Item five

1. Mechanism of Soil Acidification by Ammonium Nitrate (NH_4NO_3)

Continuous application of ammonium nitrate fertilizer increases soil acidity through:

1 **Cation Hydrolysis:** The ammonium ion (NH_4^+) undergoes hydrolysis in soil water, releasing hydronium ions (H_3O^+):

$$NH^+_{4(aq)} + H_2O_{(l)} \rightleftharpoons NH_{3(aq)} + H_3O^+_{(aq)}$$

2. Quantitative Soil Filtrate Titration & pH Analysis

- Moles of $NaOH$ neutralized in $25.0\ cm^3$:

$$\text{Moles of } NaOH = 0.05\ \text{mol dm}^{-3} \times \frac{10.0\ \text{cm}^3}{1000} = 0.0005\ \text{mol}$$

- Total Moles of Acid (H^+) in $500\ cm^3$ Soil Filtrate:

$$\text{Moles of } H^+ = 0.0005 \times \frac{500}{25.0} = \mathbf{0.01}\ \text{mol}$$

- Mass of Acid-Generating NH_4NO_3 in Soil Sample ($M_r = 80.04\ g\ mol^{-1}$):

$$\text{Mass} = 0.01\ \text{mol} \times 80.04\ \text{g mol}^{-1} = \mathbf{0.80}\ \text{g}$$

- Soil Filtrate pH Calculation:

$$[H^+] = \frac{0.01\ \text{mol}}{0.50\ \text{dm}^3} = 0.02\ \text{mol dm}^{-3}$$

$$\text{pH} = -\log_{10}(0.02) = \mathbf{1.70}$$

- Agricultural Evaluation:
  A soil pH of 1.70 indicates severe acidity, which inhibits nutrient absorption in crops requiring alkaline conditions (such as sorghum and cabbages).

3. Buffer Solution Preparation ($\mathrm{pH} = 8.95$)

To maintain the soil at $\mathrm{pH} = 8.95$, an alkaline buffer of $\mathrm{NH_3/NH_4Cl}$ is used:

1 Calculate Target pOH and $\mathrm{p}K_b$:

$$\mathrm{pOH} = 14 - \mathrm{pH} = 14 - 8.95 = 5.05$$

$$\mathrm{p}K_b = -\log_{10}(1.85 \times 10^{-5}) = 4.733$$

2 Apply the Henderson–Hasselbalch Equation for Weak Bases:

$$\mathrm{pOH} = \mathrm{p}K_b + \log_{10}\left(\frac{[\mathrm{NH_4^+}]}{[\mathrm{NH_3}]}\right)$$

$$5.05 = 4.733 + \log_{10}\left(\frac{[\mathrm{NH_4^+}]}{[\mathrm{NH_3}]}\right)$$

$$\log_{10}\left(\frac{[\mathrm{NH_4^+}]}{[\mathrm{NH_3}]}\right) = 0.317 \implies \frac{[\mathrm{NH_4Cl}]}{[\mathrm{NH_3}]} = 10^{0.317} \approx \mathbf{2.08}$$

- **Preparation Directive:** Mix ammonium chloride ($\mathrm{NH_4Cl}$) and ammonia ($\mathrm{NH_3}$) in a $2.08 : 1$ **molar ratio** to achieve and maintain a stable soil pH of 8.95.

Applications

Blood Plasma Buffering (Carbonic Acid–Bicarbonate System):

$$\mathrm{H_2CO_{3(aq)} + H_2O_{(l)} \rightleftharpoons HCO^-_{3(aq)} + H_3O^+_{(aq)}}$$

Maintains human arterial blood pH strictly between 7.35 and 7.45. Excess acid from metabolic activity is neutralized by bicarbonate ions ($\mathrm{HCO_3^-}$), while excess base reacts with carbonic acid ($\mathrm{H_2CO_3}$).

Fermentation & Brewing Operations:

Industrial alcohol production and commercial brewing require buffered culture media to keep yeast and bacterial environments at optimal pH levels for maximum enzyme activity.

Eye drops, intravenous fluids, and topical creams incorporate buffer systems (such as citrate or phosphate buffers) to match biological pH, preventing tissue irritation and stabilizing active ingredients against chemical degradation.

ASSESEMENT OBJECTIVE 1 (AO1)

1. Periodicity & Anomalies in First Ionization Energies across Period 3

- **General Trend:** First ionization energy generally increases across Period 3 (Na to Ar) due to an increasing nuclear charge (Z) and decreasing atomic radius, with shielding remaining relatively constant ($n = 3$).
- **Anomaly 1 ($\mathrm{Mg} \rightarrow \mathrm{Al}$ Dip):**
  - Mg ($1s^2 2s^2 2p^6 3s^2$) has a fully filled $3s$ subshell.
  - Al ($1s^2 2s^2 2p^6 3s^2 3p^1$) has its outermost electron in a higher-energy $3p$ orbital, which is further from the nucleus and partially shielded by the $3s$ electrons, requiring less energy to remove.
- **Anomaly 2 ($\mathrm{P} \rightarrow \mathrm{S}$ Dip):**
  - P ($1s^2 2s^2 2p^6 3s^2 3p^3$) has a stable, half-filled $3p$ subshell with unpaired electrons.
  - S ($1s^2 2s^2 2p^6 3s^2 3p^4$) contains a paired set of electrons in one $3p$ orbital. Inter-electron repulsion between paired electrons in the same orbital makes it easier to remove one electron from sulfur.

3. Industrial Applications, Hazards & Safety Mitigations

- **Applications:**
  - MgO: Used as a refractory lining for high-temperature furnaces due to its high melting point ($2852^\circ C$).
  - Al_2O_3: Primary raw material for industrial aluminum extraction (Hall–Héroult process) and used as an abrasive.
  - SiO_2: Main component in glass manufacturing, construction concrete, and ceramics.
- **Hazards & Safety Mitigations:**
  - **Severe Chemical Burns / Caustic Risks:** Solid Na_2O reacts violently with water/moisture to form corrosive $NaOH$. *Mitigation:* Store under dry inert conditions; handle using nitrile gloves, face shields, and chemical aprons.
  - **Silicosis / Respiratory Damage:** Fine dust from silica (SiO_2) causes irreversible lung fibrosis upon chronic inhalation. *Mitigation:* Install local exhaust ventilation (LEV) systems and require operators to wear certified N95/FFP3 respirators.

Item 2

1. Explanation of the Diagonal Relationship

Although Beryllium (Be) belongs to **Group II (Period 2)** and Aluminum (Al) belongs to **Group III (Period 3)**, they exhibit remarkably similar chemical and physical properties due to:

- **Similar Ionic Charge Density:** Both Be^{2+} and Al^{3+} have a high ratio of ionic charge to ionic radius ($Be^{2+}: r \approx 0.031$ nm; $Al^{3+}: r \approx 0.050$ nm).
- **Similar Electronegativity & Polarizing Power:** Both cations possess high polarizing power, distorting electron clouds of neighboring anions to impart significant covalent character to their compounds.
- **Similar Standard Oxidation Potentials:** Both metals exhibit similar ease of oxidation.

2. Chemical Reactions

a) Reactions with Dilute Hydrochloric Acid (HCl)

Both metals react with dilute hydrochloric acid to liberate hydrogen gas and form aqueous chloride salts:

$$Be_{(s)} + 2HCl_{(aq)} \rightarrow BeCl_{2(aq)} + H_{2(g)}$$

$$2Al_{(s)} + 6HCl_{(aq)} \rightarrow 2AlCl_{3(aq)} + 3H_{2(g)}$$

b) Reactions with Hot Concentrated Sodium Hydroxide (NaOH)

Because both metals are amphoteric, they dissolve in hot concentrated alkaline solutions to produce soluble complex hydroxo-aluminate/oberyllate ions and hydrogen gas:

$$Be_{(s)} + 2NaOH_{(aq)} + 2H_2O_{(l)} \rightarrow Na_2[Be(OH)_4]_{(aq)} + H_{2(g)}$$

$$2Al_{(s)} + 2NaOH_{(aq)} + 6H_2O_{(l)} \rightarrow 2Na[Al(OH)_4]_{(aq)} + 3H_{2(g)}$$

Item three

3. Industrial & Technological Applications

- **Beryllium (Be):**
  - **Aerospace & Defence:** Used in structural components for aircraft, satellites, and spacecraft due to its high stiffness-to-weight ratio and dimensional stability.
  - **Nuclear Reactors:** Serves as a neutron reflector and moderator because of its low neutron absorption cross-section.
  - **Copper-Beryllium Alloys:** Used in non-sparking tools for flammable environments, electrical contacts, and precision springs.
- **Aluminum (Al):**
  - **Transportation & Structural Engineering:** Lightweight structural material for automotive bodies, railway cars, and aviation.
  - **Electrical Power Distribution:** Used in overhead high-voltage transmission lines (ACSR cables) due to high conductivity and light weight.
  - **Packaging & Construction:** Used in beverage cans, food foils, and architectural window frames.

4. Associated Hazards & Health/Environmental Risks

- **Beryllium Hazards:**
  - **Berylliosis (Chronic Beryllium Disease - CBD):** Inhalation of beryllium dust or soluble salts causes an irreversible pulmonary autoimmune condition marked by lung granulomas.
  - **Carcinogenicity:** Beryllium and its compounds are classified as Class 1 human carcinogens.
- **Aluminum Hazards:**
  - **Neurotoxicity:** Excessive systemic accumulation of aluminum ions (Al^{3+}) is linked to neurodegenerative disorders and dialysis encephalopathy.
  - **Ecosystem Acidification:** Acid rain leaches Al^{3+} into aquatic bodies, binding to fish gills and causing severe osmoregulatory distress.

5. Safety Mitigations & Control Measures

1. **Engineering Controls:** Install closed containment systems, local exhaust ventilation (LEV), and high-efficiency particulate air ($HEPA$) filtration units wherever beryllium or aluminum dust/fumes are generated.
2. **Personal Protective Equipment (PPE):** Operators handling beryllium or fine aluminum powders must wear certified full-face air-purifying respirators (P3/N100), protective suits, and nitrile gloves.
3. **Medical Surveillance & Exposure Monitoring:** Implement regular spirometry testing and air sampling to keep occupational exposure limits (OEL) below regulated thresholds.
4. **Effluent Neutralization:** Treat industrial runoff with alkaline precipitating agents to capture Al^{3+} as insoluble hydroxides before environmental discharge.

Item three

1. Physical Characteristics of Radioactive Materials and Types of Radiation

Radioactive materials consist of unstable atomic nuclei that spontaneously undergo radioactive decay to achieve greater stability by emitting radiation:

- **Alpha (α) Particles:**
  - **Composition:** Helium nuclei ($^4_2\mathrm{He}^{2+}$) consisting of 2 protons and 2 neutrons.
  - **Deflection in Electric Field:** Deflects **towards the negative plate** due to its positive charge ($+2e$).
  - **Penetrating/Ionizing Power:** Highly ionizing due to large mass and charge; low penetrating power (stopped by a sheet of paper or human skin).
- **Beta (β) Particles:**
  - **Composition:** High-energy, high-speed electrons ($^0_{-1}\mathrm{e}$ or β^-) emitted from the nucleus when a neutron decays into a proton.
  - **Deflection in Electric Field:** Deflects **strongly towards the positive plate** due to its negative charge ($-1e$) and low mass.
  - **Penetrating/Ionizing Power:** Moderately ionizing; moderate penetrating power (stopped by a few millimeters of aluminum).
- **Gamma (γ) Rays:**
  - **Composition:** High-energy electromagnetic radiation ($^0_0\gamma$) with zero mass and zero charge.
  - **Deflection in Electric Field: Undeflected** (passes straight through).
  - **Penetrating/Ionizing Power:** Weakly ionizing; extremely high penetrating power (requires dense lead or thick concrete shielding to attenuate).

2. Decay Sequence of Lead-212 ($^{212}_{82}\mathrm{Pb}$)

The decay process consists of three sequential decay steps:

1 **Beta Decay (β^-):**

$$^{212}_{82}\mathrm{Pb} \longrightarrow {}^{212}_{83}\mathrm{Bi} + {}^{0}_{-1}\mathrm{e}$$

Nuclide formed: **Bismuth-212 ($^{212}_{83}\mathrm{Bi}$)**

2 **Alpha Decay (α):**

$$^{212}_{83}\mathrm{Bi} \longrightarrow {}^{208}_{81}\mathrm{Tl} + {}^{4}_{2}\mathrm{He}$$

Nuclide formed: **Thallium-208 ($^{208}_{81}\mathrm{Tl}$)**

3 **Beta Decay (β^-):**

$$^{208}_{81}\mathrm{Tl} \longrightarrow {}^{208}_{82}\mathrm{Pb} + {}^{0}_{-1}\mathrm{e}$$

Nuclide formed: **Lead-208 ($^{208}_{82}\mathrm{Pb}$)**

3. Final Stable Isotope in Thorium-232 ($^{232}_{90}\mathrm{Th}$) Decay Series

Thorium-232 naturally decays by releasing **4 beta particles** ($4 \times {}^{0}_{-1}\mathrm{e}$) and **6 alpha particles** ($6 \times {}^{4}_{2}\mathrm{He}$).

- **Mass Number (A):**

$$A = 232 - (6 \times 4) - (4 \times 0) = 232 - 24 = 208$$

- **Atomic Number (Z):**

$$Z = 90 - (6 \times 2) - (4 \times -1) = 90 - 12 + 4 = 82$$

- **Identification:**
  The element with atomic number $Z = 82$ is Lead (Pb).
- **Final Nuclear Reaction:**

$$^{232}_{90}\mathrm{Th} \longrightarrow {}^{208}_{82}\mathrm{Pb} + 6{}^{4}_{2}\mathrm{He} + 4{}^{0}_{-1}\mathrm{e}$$

- **Final Stable Isotope:** Lead-208 ($^{208}_{82}\mathrm{Pb}$).

4. Residual Activity Calculation for Experiment Four

- Given:
  - Initial activity (A_0) = 1000 counts s^{-1}
  - Half-life ($t_{1/2}$) = 3 hours
  - Elapsed time (t) = 2 hours
- Step 1: Calculate the Decay Constant (λ)

$$\lambda = \frac{\ln(2)}{t_{1/2}} = \frac{0.69315}{3 \text{ hours}} \approx 0.23105 \text{ h}^{-1}$$

- Step 2: Calculate Residual Activity (A) using Radioactive Decay Law

$$A = A_0 \cdot e^{-\lambda t}$$

$$A = 1000 \cdot e^{-(0.23105 \times 2)}$$

$$A = 1000 \cdot e^{-0.4621}$$

$$A = 1000 \times 0.6300 = \mathbf{630} \text{ counts } \mathbf{s^{-1}}$$

5. Real-World Health Risks & Safety Protocols

- Health Hazards:
  - Cellular & DNA Damage: Ionizing radiation breaks chemical bonds in biological tissue, causing double-strand DNA breaks that lead to cell death or mutations.
  - Somatic Effects: Acute exposure causes radiation sickness (nausea, hair loss, tissue necrosis), while chronic low-dose exposure increases long-term cancer risks (e.g., leukemia, thyroid cancer).
  - Genetic Effects: Mutations in germ cells passed down to offspring.
- Safety & Control Measures:
  - Shielding: Enclose radioactive reactors and waste in thick lead and high-density reinforced concrete containment structures.
  - Distance & Remote Handling: Utilize telemanipulators and robotic arms to handle radioactive substances; radiation intensity decreases exponentially with distance ($I \propto 1/r^2$).
  - Personal Dosimetry & Containment: Staff must wear personal dosimeter badges (TLDs) to monitor exposure limits, alongside full protective hazmat suits with personal respirators.

Item Four

1. Physical Properties & Trends in Physical States

Under room temperature and pressure ($25^\circ C$, 1 atm), the halogens display a gradual transition in physical state down Group 17:

- Fluorine (F_2): Pale yellow gas
- Chlorine (Cl_2): Yellow-green gas
- Bromine (Br_2): Dense, volatile dark-red/red-brown liquid
- Iodine (I_2): Dark grey/black shiny crystalline solid that sublimes readily to form a purple vapor

Fluorine and chlorine are gases, bromine is a volatile red brown or dark red heavy liquid and iodine is a grey black solid at room temperature. The molecules are held together by weak Van der Waal's forces of attraction which increase in magnitude from fluorine molecule to iodine molecule because of increase in molecular weight, atomic radius of the atoms forming the molecule and total number of electrons in the molecule. These forces are sufficiently strong to bind the Iodine molecules into a three-dimensional solid crystalline lattice. Iodine sublimes because these forces are weak and are easily broken down on heating.

2. Chemical Reactions of Halogens

a) Reactions with Water (H_2O)

- Fluorine (F_2): Oxidizes water rapidly and exothermically, releasing oxygen gas:

$$2F_{2(g)} + 2H_2O_{(l)} \rightarrow 4HF_{(aq)} + O_{2(g)}$$

- Chlorine (Cl_2): Undergoes disproportionation (simultaneous oxidation and reduction) to form hydrochloric acid and chloric(I) acid (hypochlorous acid):

$$Cl_{2(g)} + H_2O_{(l)} \rightleftharpoons HCl_{(aq)} + HClO_{(aq)}$$

- Bromine (Br_2): Disproportionates slightly in water to give a weak reddish-yellow solution of hydrobromic acid and bromic(I) acid:

$$Br_{2(l)} + H_2O_{(l)} \rightleftharpoons HBr_{(aq)} + HBrO_{(aq)}$$

- Iodine (I_2): Practically insoluble in pure water; does not react appreciably due to its low oxidizing power ($E^\circ_{I_2/I^-} = +0.54$ V).

b) Reactions with Sodium Hydroxide ($NaOH$)

- **Cold, Dilute $NaOH$ Solution ($15°C$):**
  Chlorine, bromine, and iodine disproportionate to form halide and halate(I) ions:

$$X_2 + 2NaOH_{(aq)} \rightarrow NaX_{(aq)} + NaXO_{(aq)} + H_2O_{(l)} \quad (\text{where X =}$$

Example with Chlorine:

$$Cl_{2(g)} + 2NaOH_{(aq)} \rightarrow NaCl_{(aq)} + NaClO_{(aq)} + H_2O_{(l)}$$

(Forming sodium chlorate(I), the active ingredient in commercial household bleach).

- **Hot, Concentrated $NaOH$ Solution ($70°C$):**
  Halides disproportionate further to form halide and halate(V) ions:

$$3X_2 + 6NaOH_{(aq)} \rightarrow 5NaX_{(aq)} + NaXO_{3(aq)} + 3H_2O_{(l)}$$

Example with Chlorine:

$$3Cl_{2(g)} + 6NaOH_{(aq)} \rightarrow 5NaCl_{(aq)} + NaClO_{3(aq)} + 3H_2O_{(l)}$$

(Forming sodium chlorate(V), used as a weedkiller and oxidizing agent).

3. Industrial, Domestic, and Medical Applications

- **Chlorine (Cl_2):** Water purification and municipal sewage treatment (killing waterborne pathogens via $HClO$ action); synthesis of polyvinyl chloride (PVC) plastics and household bleaches ($NaClO$).
- **Fluorine (F_2):** Manufacture of fluoropolymers (such as Teflon / $PTFE$), hydrofluorocarbon refrigerants, and sodium fluoride (NaF) additives in toothpaste to prevent dental caries.
- **Bromine (Br_2):** Production of flame retardants ($BFRs$) added to electronics and textiles, agricultural pesticides (methyl bromide alternatives), and pharmaceuticals.
- **Iodine (I_2):** Medical antiseptics and disinfectants (povidone-iodine tincture); essential dietary constituent added to table salt (KIO_3) to prevent goiter and thyroid deficiency.

4. Associated Health/Environmental Hazards & Safety Mitigations

- Hazards:
  - **Acute Toxicity & Respiratory Damage:** Chlorine and fluorine gases are potent choking agents that irritate mucous membranes, causing severe pulmonary edema and chemical burns upon inhalation.
  - **Corrosive Tissue Damage:** Liquid bromine causes severe, slow-healing chemical burns and deep skin ulcers upon contact.
  - **Environmental Bioaccumulation:** Halon gas and halogenated hydrocarbons (such as CFCs) cause stratospheric ozone layer depletion.
- Safety Mitigations:
  - **Fume Hood & Extraction Isolation:** Perform all handling and synthesis involving free halogens inside high-efficiency fume cupboards equipped with gas scrubber systems (using aqueous NaOH or $\mathrm{Na_2S_2O_3}$).
  - **Personal Protective Equipment (PPE):** Operators must wear full-face gas respirators with halogen-specific filter cartridges, butyl rubber heavy-duty gloves, and chemical splash aprons.
  - **Spill Neutralization Protocols:** Neutralize accidental bromine or chlorine spills immediately using dilute sodium thiosulfate ($\mathrm{Na_2S_2O_3}$) or sodium hydrogen carbonate ($\mathrm{NaHCO_3}$) solution.

ASSESEMENT OBJECTIVE 2 (AO2)

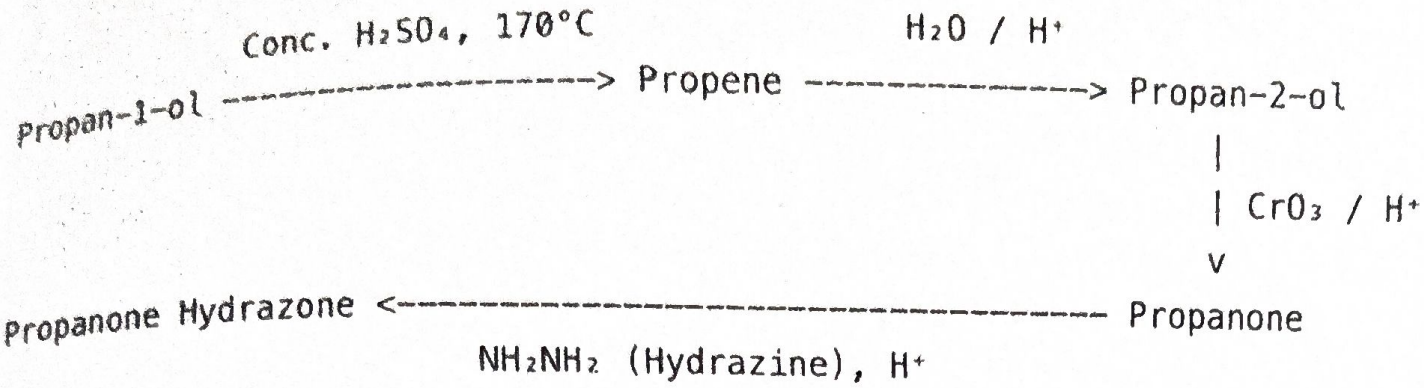
Propan-1-ol is a liquid with a relatively high boiling point because in addition to the Van der Waals forces of attraction, the molecules associate through hydrogen bonding.

Propan-1-ol is completely miscible with water because it is polar and its molecules associate with water molecules through hydrogen bonding.

Benzene is a which boils boils at a high temperature though less than that of propanol because it molecules associate through Van der Waal forces whose magnitude increase with increased molecular mass. Benzene has a high molecular mass, so the Van der Waal's forces are strong enough to make it a liquid.

Benzene is Insoluble in water due because its non-polar, but miscible with non-polar organic solvents.

Synthesis of propan-1-ol to propanone hydrazine



Conversion of Benzene to N-Ethyl-N-phenylamine

1 **Step 1: Nitration of Benzene**

- **Reagents & Conditions:** Concentrated nitric acid (HNO_3) and concentrated sulfuric acid (H_2SO_4) at 50 °C–60 °C.
- **Equation:**

$$C_6H_6 + HNO_3 \xrightarrow{\text{conc. } H_2SO_4,\ 55^\circ C} C_6H_5NO_2 + H_2O$$

2 **Step 2: Reduction of Nitrobenzene to Phenylamine (Aniline)**

- **Reagents & Conditions:** Tin (Sn) or Iron (Fe) in concentrated hydrochloric acid (HCl), followed by addition of aqueous $NaOH$.
- **Equation:**

$$C_6H_5NO_2 + 6[H] \xrightarrow{Sn/HCl,\ \text{then } OH^-} C_6H_5NH_2 + 2H_2O$$

3 **Step 3: Mono-alkylation to N-Ethylphenylamine**

- **Reagents & Conditions:** Haloethane (CH_3CH_2Br) under controlled heating.
- **Equation:**

$$C_6H_5NH_2 + CH_3CH_2Br \rightarrow C_6H_5NHCH_2CH_3 + HBr$$

Chemical Component	Associated Hazards	Risk Mitigation & Management Strategies
Benzene	Carcinogen (causes leukemia), highly flammable, volatile, hazardous to aquatic life.	Use sealed automated transfer systems, work in fume hoods, monitor workplace vapor thresholds, and enforce use of organic vapor respirators.
Propan-1-ol	Flammable liquid and vapor, causes serious eye damage, causes central nervous system depression.	Keep away from open flames/heat sources, store in explosion-proof cabinets, and mandate splash goggles and nitrile gloves.